Cover Song Generation

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About me

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Research Topics

Music generation & arrangement
Cover song generation

Education

2023–present
Ph.D.
Electrical Engineering

2013–2018, 2020–2022
B.S. & M.S.
Computer Science

Experience

2026–present
Moises
Research Collaboration

2025–present
MCT Lab
Research Assistant

2023–2025
KKCompany (KKBOX)
Research Engineer

2022–2023
Academia Sinica
Research Assistant

Have you ever heard a song that sounds so similar to another song?

Example

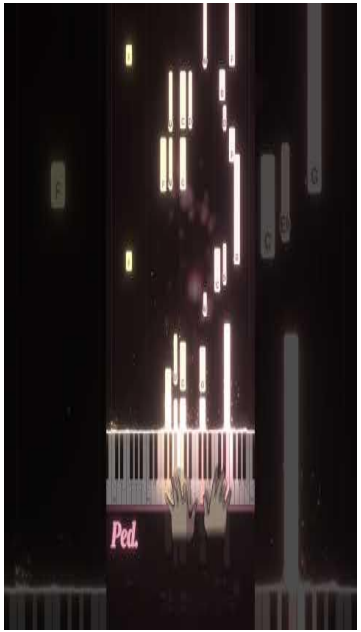
- Laufey - From the Start

Same

Melody
Chords

Changed

Instruments
Meter



Song adaptation

Arrangement

“In music, an arrangement refers to the way in which a composer or arranger takes an existing musical composition and reinterprets it, altering various elements such as the **harmony, rhythm, texture, tempo, dynamics, or instrumentation**.”

Adaptation

“A new version of a pre-existing composition that changes **lyrics, language, structure, or style beyond mere arrangement**.”

Arrangement: https://www.aftersunsetmusic.com/resources/glossary/arrangement/
Adaptation: https://www.aftersunsetmusic.com/resources/glossary/adaptation/

Analytic view

Musical Version Identification (VI)

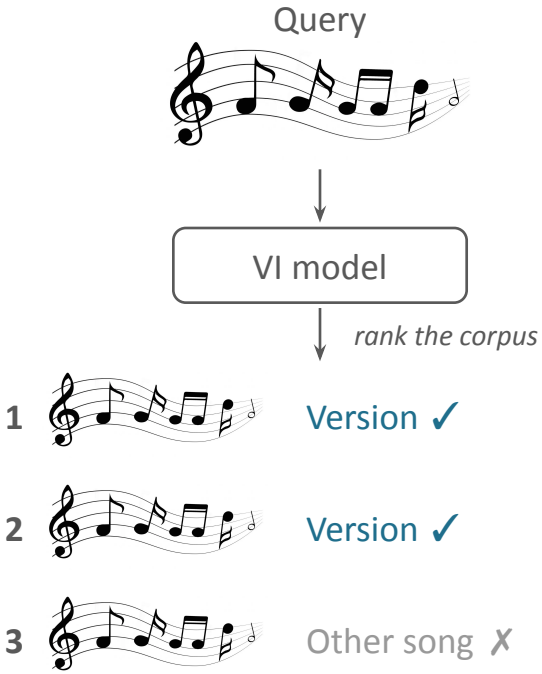
What is VI?

Task: given a **query** track, retrieve **all its versions** from a **corpus**.

- Versions of the same song form a **clique**.
- Challenge: be **invariant** to key, tempo, structure, and timbre, but keep **melody and harmony**.

Evaluation: **retrieval** metrics (MAP, MR1, P@K).

Applications: copyright detection, catalog organization, music discovery.



Yesiler et al. “Audio-based Musical Version Identification: Elements and Challenges” in IEEE Signal Processing Magazine 2021

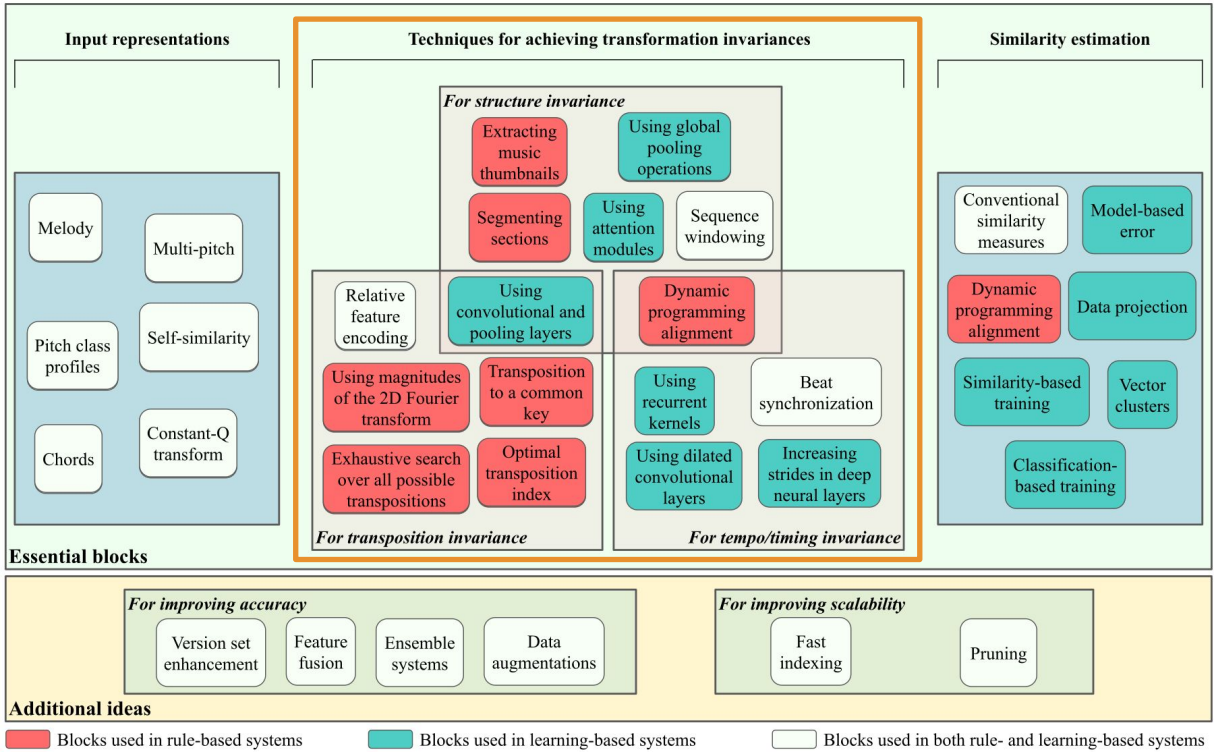
Types of musical versions

Musical Characteristic	Version Type																	
	Duplicate	Remaster	Radio Edit	Translation	Performance	Demo	Parody	Within-Genre	Karaoke	Live	Standard	Mashup	Acoustic	Medley	Remix	Cross-Genre	Arrangement	Quotation
Melody	0	0	0	0	0	1	0	1	2	1	1	0	1	1	1	2	2	2
Harmony	0	0	0	0	0	1	0	1	0	0	2	0	1	1	2	2	2	3
Tempo	0	0	0	0	2	1	1	1	0	2	1	3	2	2	3	2	2	3
Timing	0	0	0	0	2	1	1	1	0	2	1	3	2	2	2	3	3	3
Structure	0	0	1	0	1	1	1	1	1	2	2	3	2	3	3	2	3	3
Lyrics	0	0	1	3	0	1	3	0	3	1	0	0	0	1	1	1	1	2
Key	0	0	0	0	1	1	1	1	1	1	1	2	2	2	2	3	3	3
Timbre	0	0	0	0	1	1	1	1	2	2	3	2	3	2	3	3	3	3
Noise	0	1	1	1	3	3	2	3	2	3	3	2	3	3	2	3	3	3

Degree of Potential Difference	0	1	2	3
	Likely the same	May be variations	May be major differences	May be unrelated

Yesiler et al. “Audio-based Musical Version Identification: Elements and Challenges” in IEEE Signal Processing Magazine 2021

VI methods



Transformation invariances

Transposition

Different key: common-key transposition, OTI, 2D Fourier magnitude, CNN pooling

Tempo / timing

Different speed or timing: beat sync, DP alignment, dilated convolutions

Structure

Added, skipped, or repeated sections: thumbnails, segmentation, global pooling, attention

Yesiler et al. “Audio-based Musical Version Identification: Elements and Challenges” in IEEE Signal Processing Magazine 2021

VI datasets

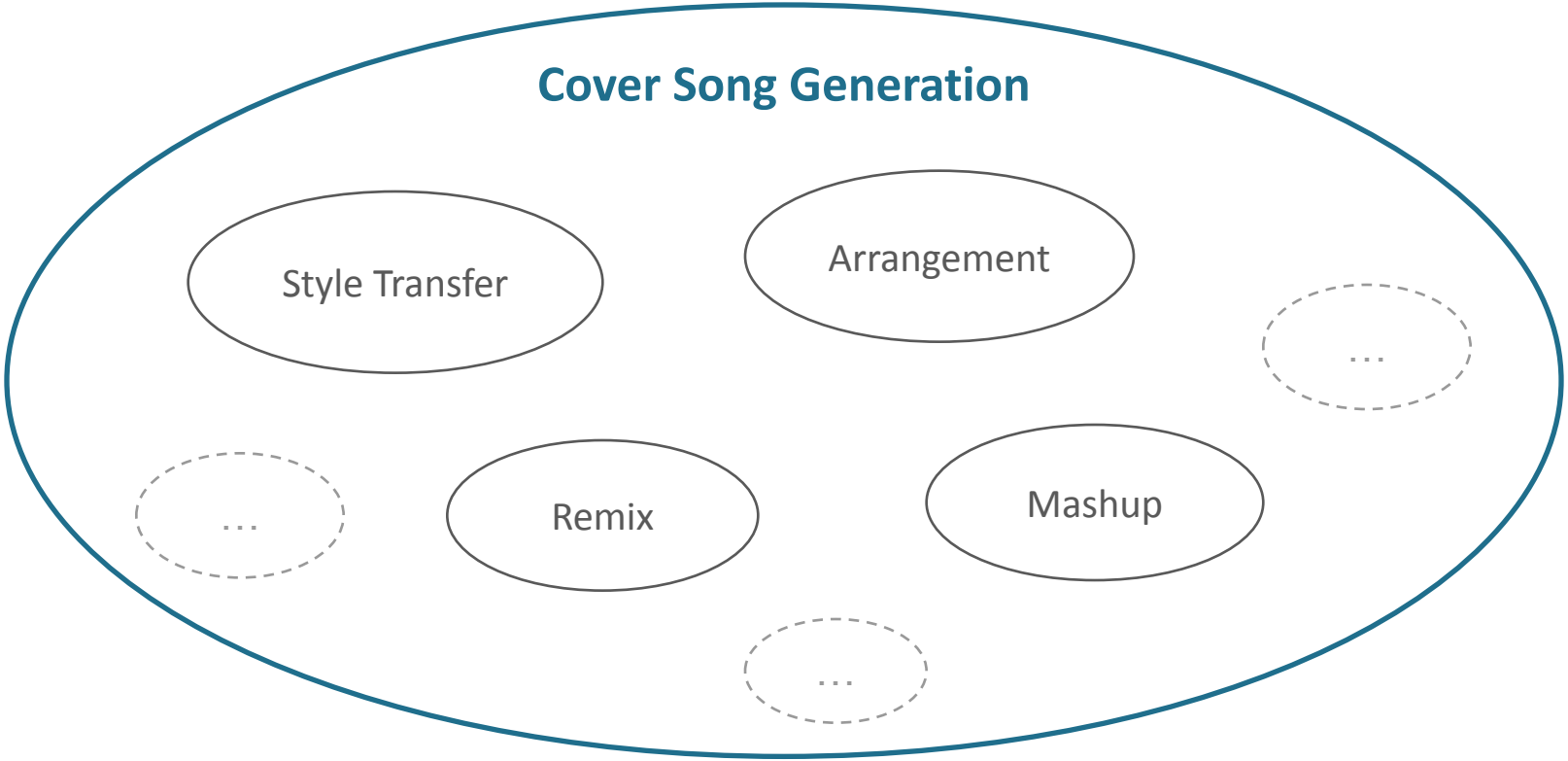
Dataset	Tracks (cliques)	What is released
MIREX (2006)	1,000 (30), incl. 670 noise tracks	Private collection
covers80 (2007)	160 (80)	Full audio + metadata
SecondHandSongs / MSD (2011)	18,196 (5,854)	Pre-extracted features + metadata
YouTubeCovers (2015)	350 (50)	Pre-extracted PCP features + metadata
SHS100K (2018)	108,869 (9,202)	YouTube URLs + metadata
Da-TACOS (2019)	15,000 (1,000) benchmark + training set	Pre-extracted features + metadata
SHS5+ / SHS4- (2019)	110,794 (26,905)	Pre-extracted features (CQT, melody, PCP)

Yesiler et al. “Audio-based Musical Version Identification: Elements and Challenges” in IEEE Signal Processing Magazine 2021

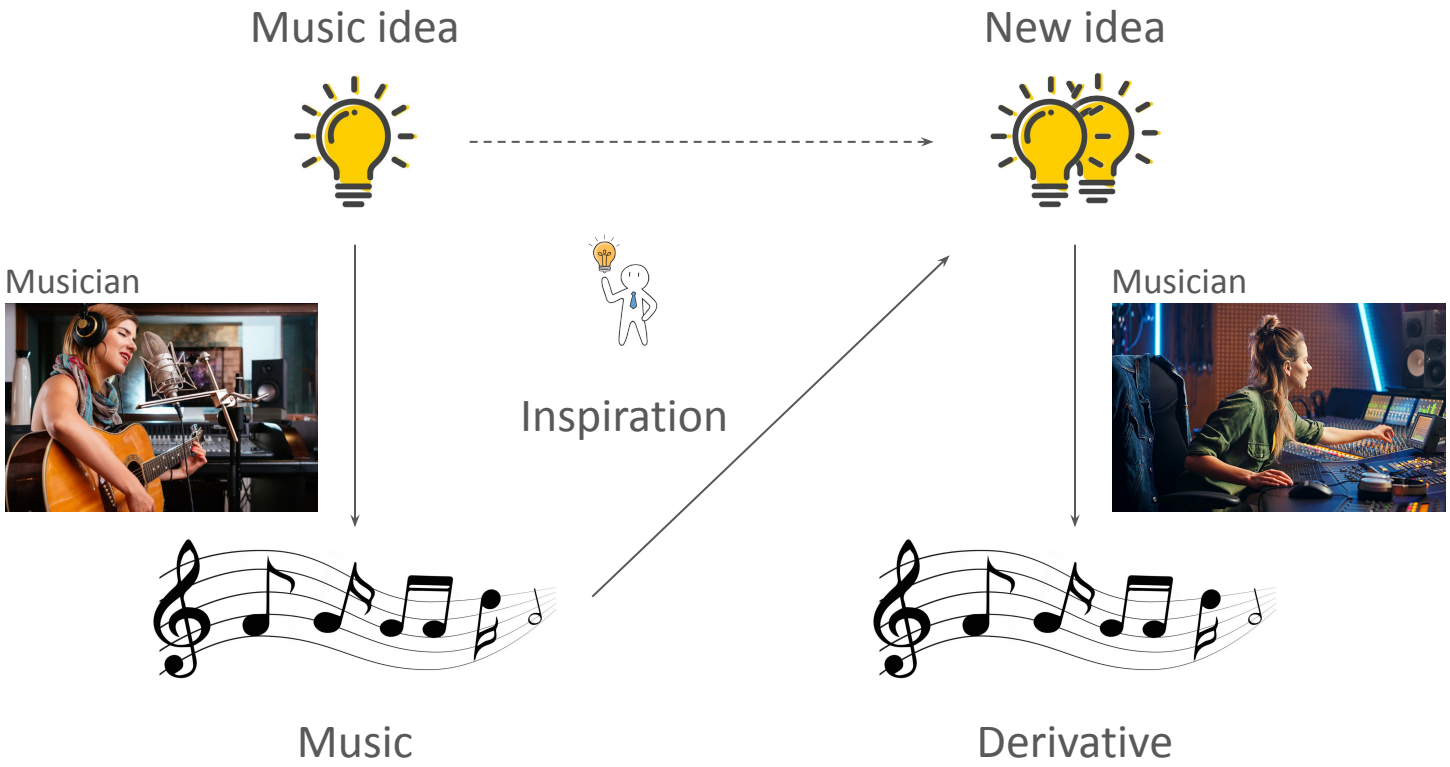
Generative view

Cover Song Generation

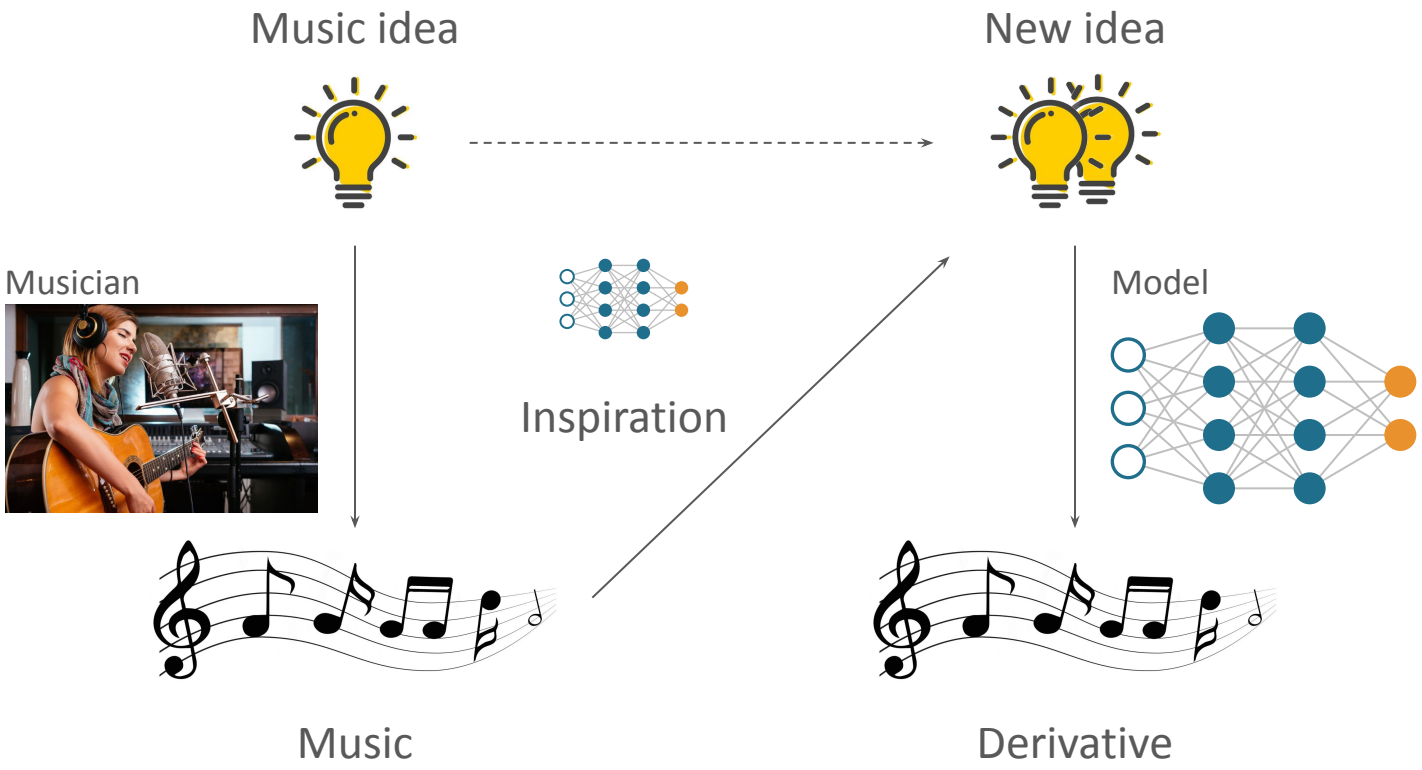
Types of cover song generation



Create a new version



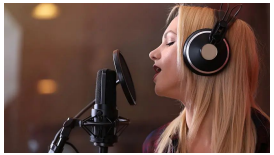
Create a new version with AI



Music production workflow

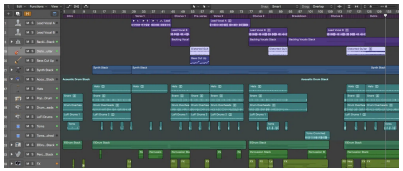
The process of creating a final musical piece involves a sequence of creative and technical stages, transforming an initial idea into a polished, publishable track.

Compose



Record

Publish



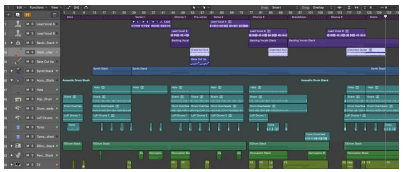
Arrange

Mix

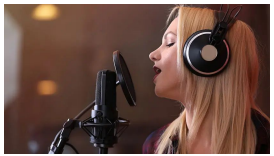
Music adaptation workflow

This workflow represents the creative process of developing a new piece of music ***inspired by or derived from*** existing material, rather than starting from an original composition.

Idea from other music



Arrange



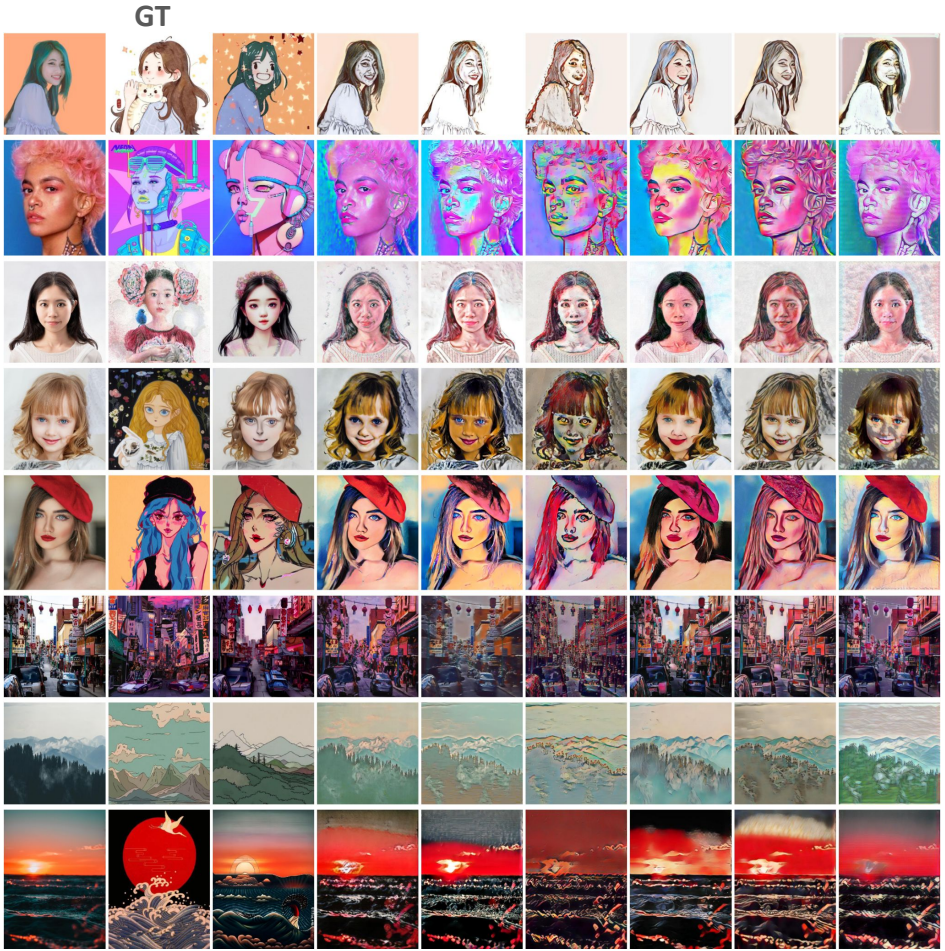
Record

Mix

Publish

Image style transfer

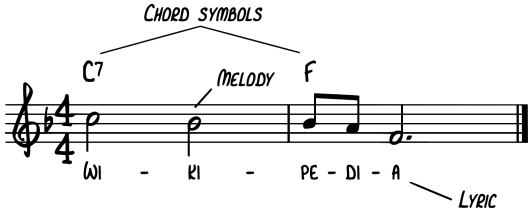
We take the **edges** of the content image and dress them up in the texture of the style image.



Zhang et al. “Inversion-Based Style Transfer with Diffusion Models” in Proc. of CVPR 2023

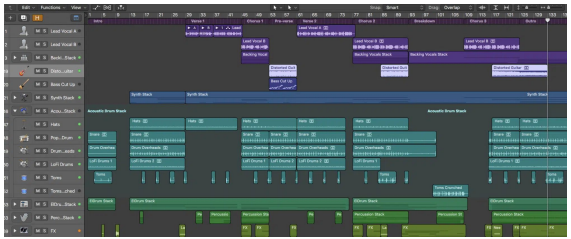
Music style transfer

- Content
  - Melody
  - Chords
- Style
  - Everything except content



Genre

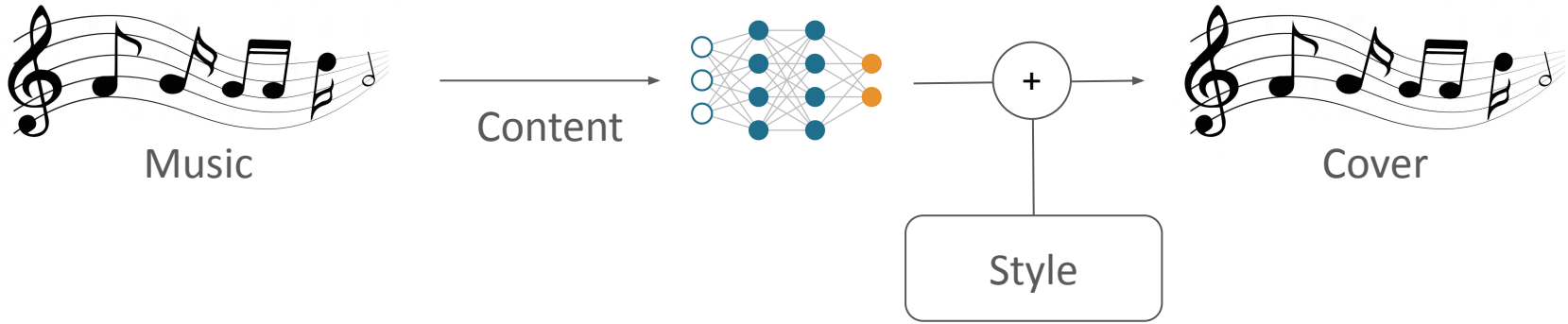
Instrument



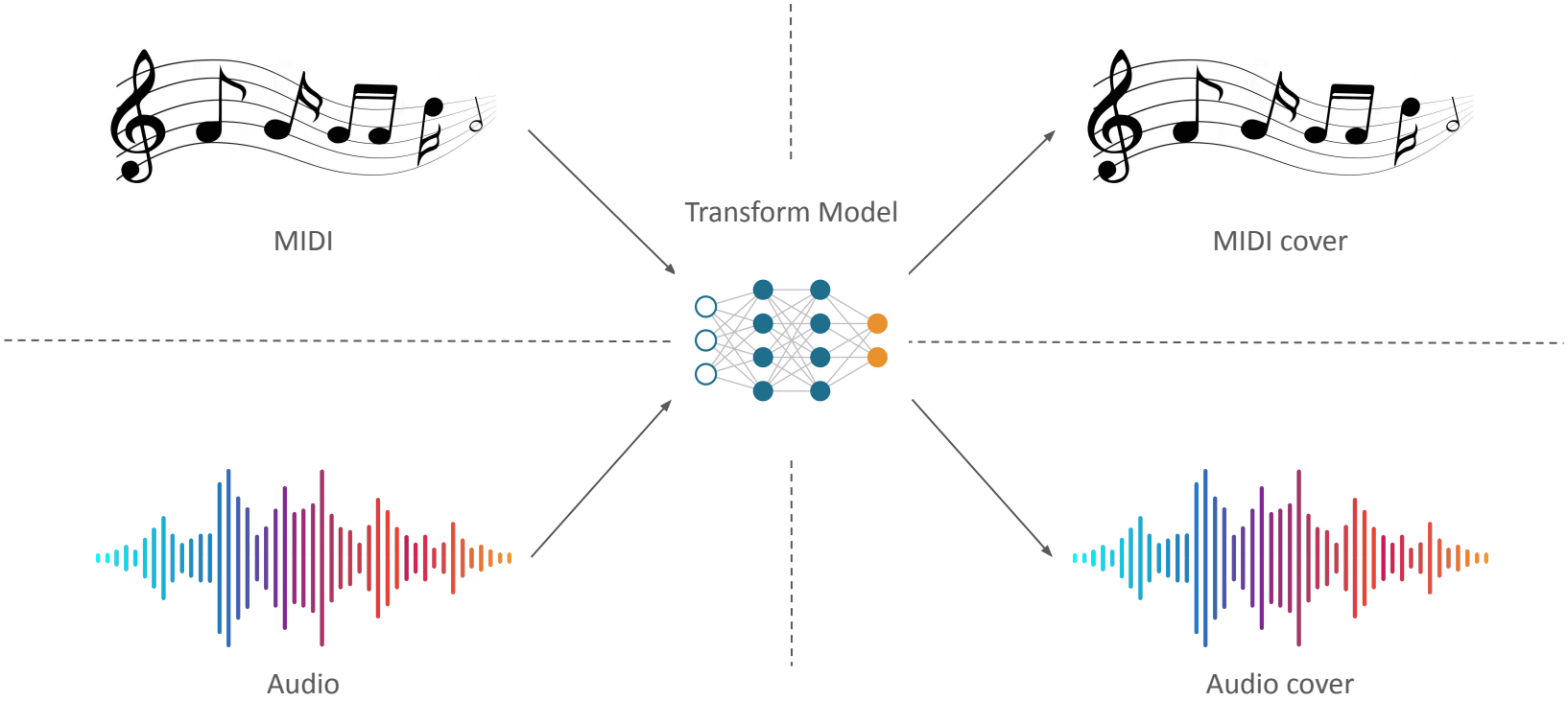
Arrangement

Problem formulation

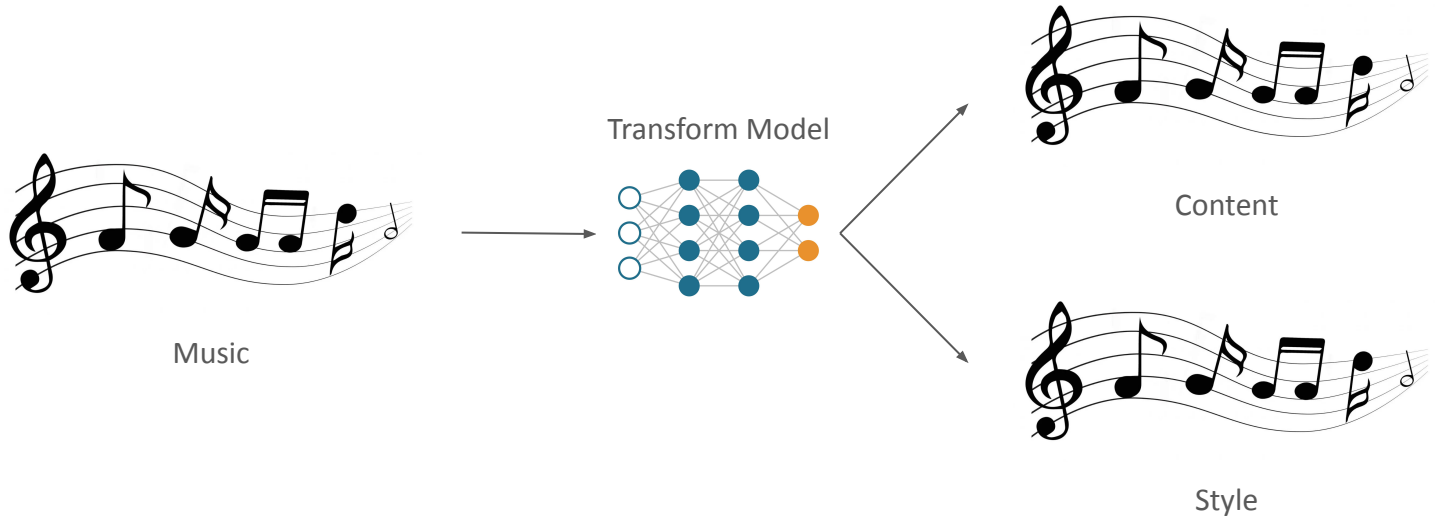
- Cover Song Generation is formulated as a problem of ***Content-Preserving Style Transfer***
  - Extracting the content (core **melody**, **harmony**, and **structure**) from the source track.
  - Replacing or re-rendering the expressive and temporal style (rhythm, timbre, articulation, dynamics) of the content with a target style (e.g., genre or artist).



Cross-modal music transformation

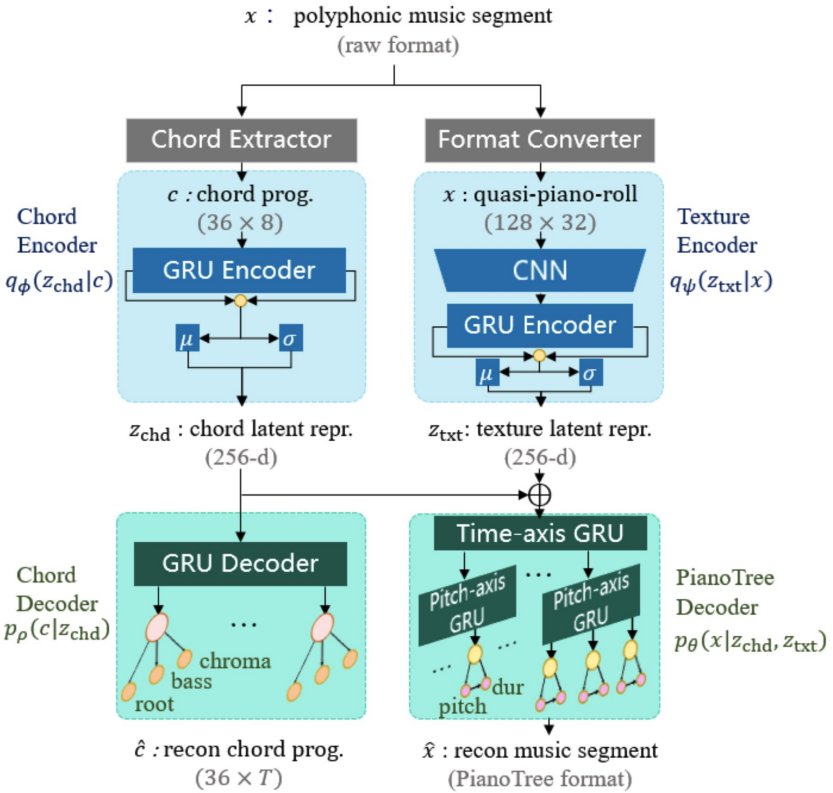


Task: disentangle music content



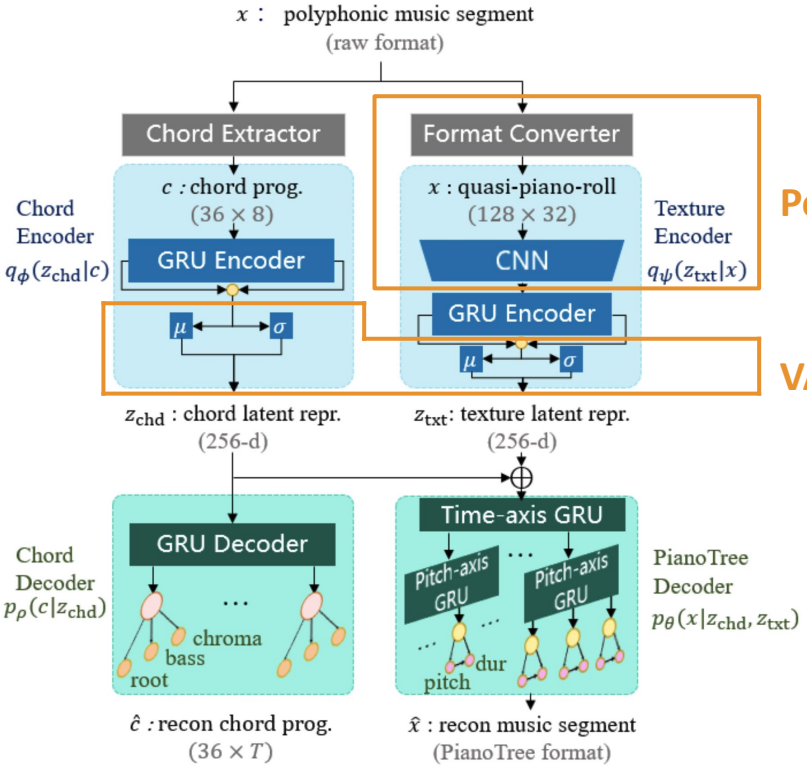
Controllable polyphonic music generation

- Disentangle chords (content) from music
  - Melody is not considered content



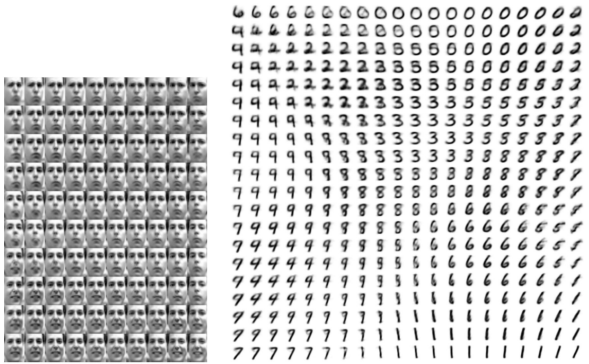
Wang et al. “Learning Interpretable Representation for Controllable Polyphonic Music Generation” in Proc. of ISMIR 2020

Controllable polyphonic music generation



Pooling

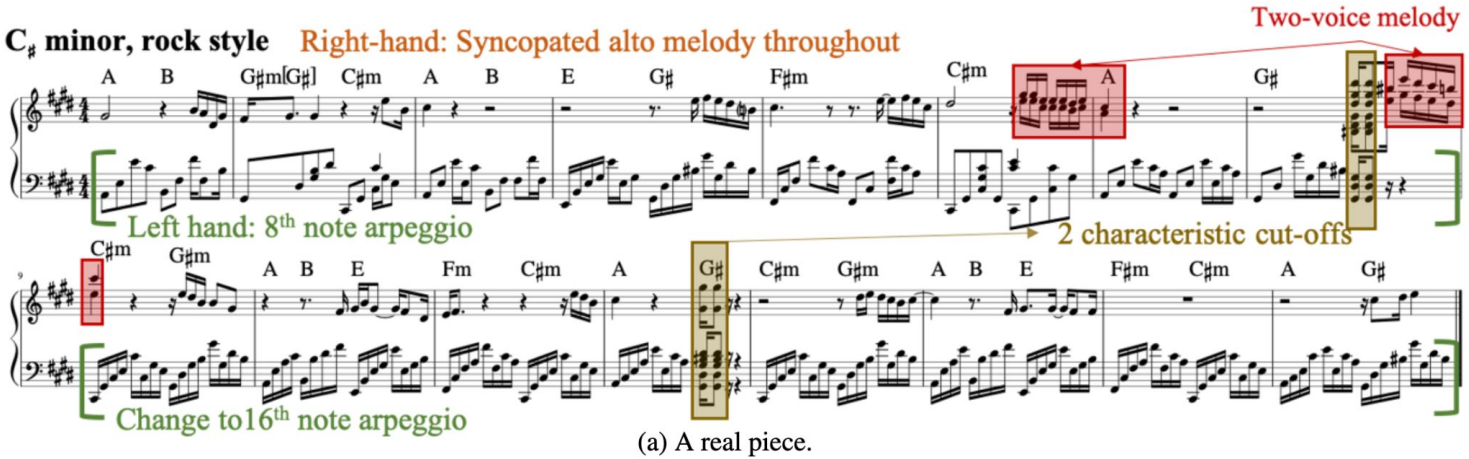
VAE



Wang et al. “Learning Interpretable Representation for Controllable Polyphonic Music Generation” in Proc. of ISMIR 2020
Kingma and Welling. “Auto-Encoding Variational Bayes” in Proc. of ICLR 2014

Controllable polyphonic music generation

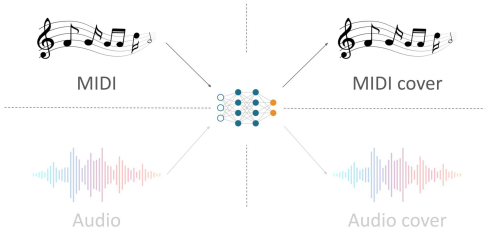
However, the melodic fidelity of the generated output is poor.



(a) A real piece.

Wang et al. “Learning Interpretable Representation for Controllable Polyphonic Music Generation” in Proc. of ISMIR 2020

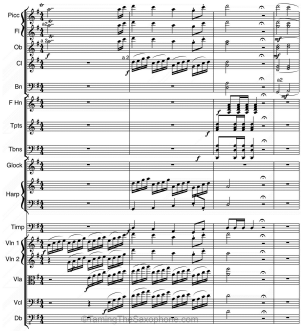
Task: arrangement with provided content



MIDI to MIDI

Melody + Chords

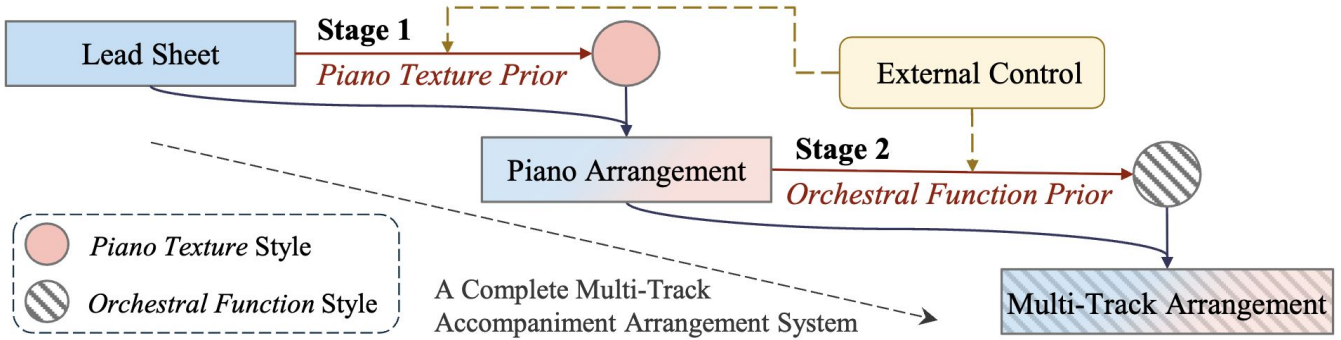
Transform Model



Arrangement

Multi-track accompaniment arrangement

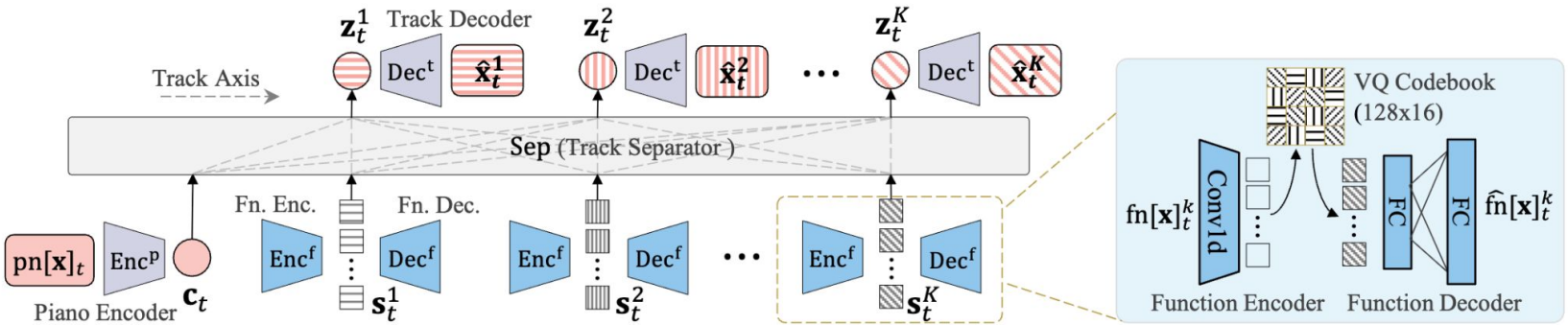
Concept 1: Arrange multi-track accompaniments from a simple lead sheet.



Zhao et al. "Structured Multi-Track Accompaniment Arrangement via Style Prior Modelling" in Proc. of NeurIPS 2024

Multi-track accompaniment arrangement

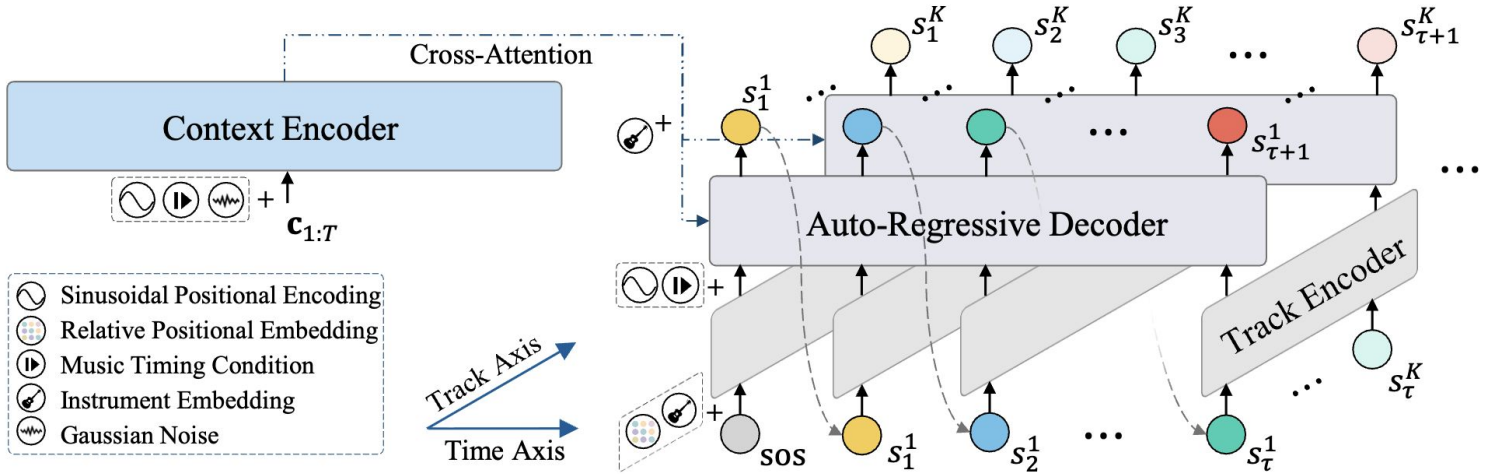
Concept 2: Generate each track by deriving the musical content from the **reduced piano** arrangement and assigning its instrumental role using a **quantized orchestral function**.



Zhao et al. “Structured Multi-Track Accompaniment Arrangement via Style Prior Modelling” in Proc. of NeurIPS 2024

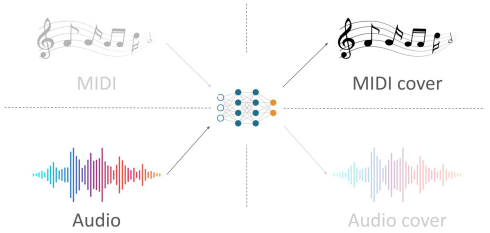
Multi-track accompaniment arrangement

Model: Generate a complete orchestral score via the automatic orchestration of a piano reduction.



Zhao et al. “Structured Multi-Track Accompaniment Arrangement via Style Prior Modelling” in Proc. of NeurIPS 2024

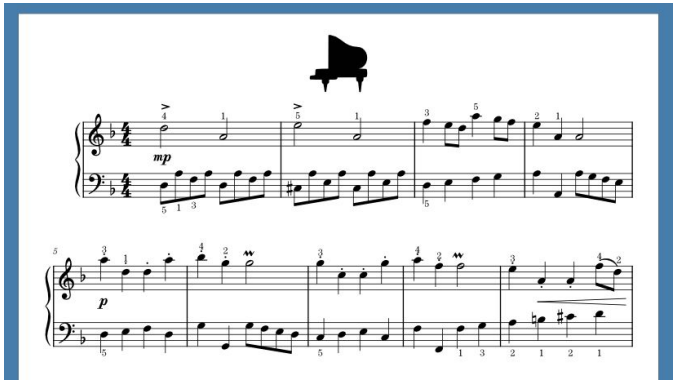
Task: piano arrangement from audio



Audio to MIDI

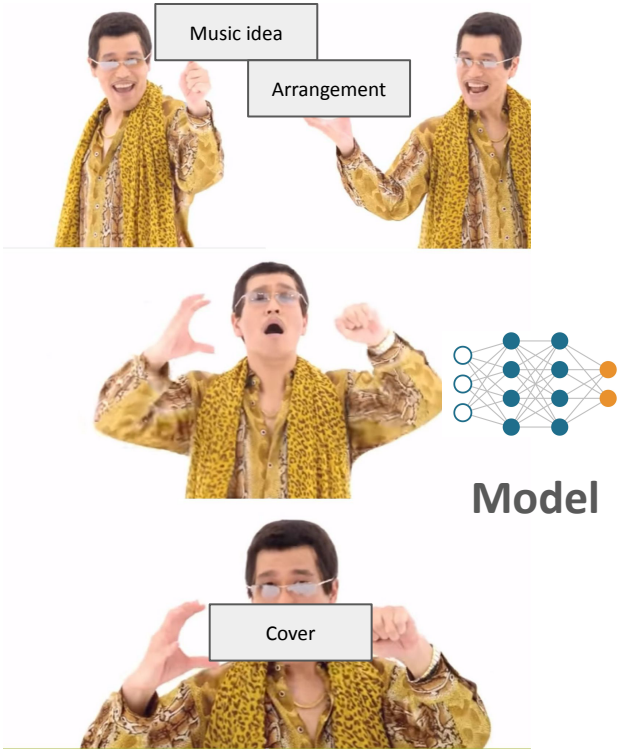
Song audio

Transform Model



Arrangement

Rethink: cover generation = extraction + generation



Content preservation

The Question: Should the shared content be preserved exactly when generating a cover?

- No. We aim for **perceptual similarity**, not strict note-for-note identity. The content should be **adaptable** to the new style.

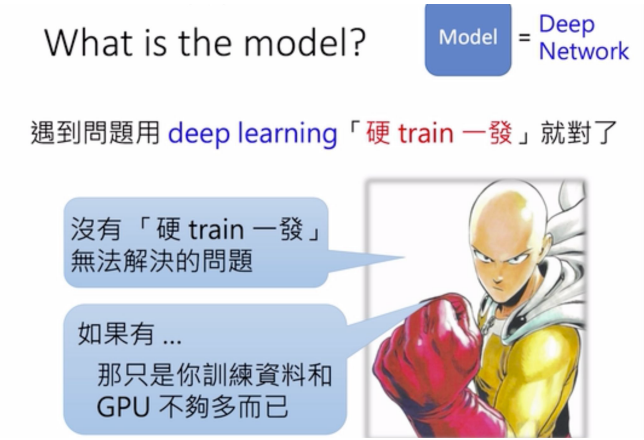
Preservation of content and style

The Challenge of Content: to what extent must the **original melody** be altered (or preserved) to fit the target style?

- Specifically: how do we define the "**essential content**" of the melody?
- More precisely: which notes are **mandatory** for recognition, and which can be changed for embellishment?

The Challenge of Style: beyond the core melody and harmony, which **other musical attributes** should be preserved in the cover song?

Further reading: Dai et al. “Music Style Transfer: A Position Paper” in Proc. of MUME 2018



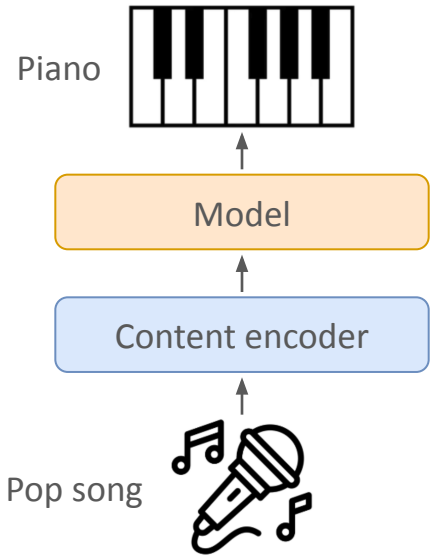
How about data-driven approaches?

"..... 不知道什麼是'硬train一發' 的，我試著解釋一下什麼是'硬train一發'。這個，'硬train一發' 啊...它是一個非常神奇的東西：它是一種***信念***，它是一種***夢想***，它是一種***浪漫***，它是***人類最原始的衝動***，它是***亙古以來人類的目標*** ....." - by 李宏毅

遇到问题就用Deep Learning【硬train一发】就对了：https://www.bilibili.com/video/BV1kE411A7Uf/

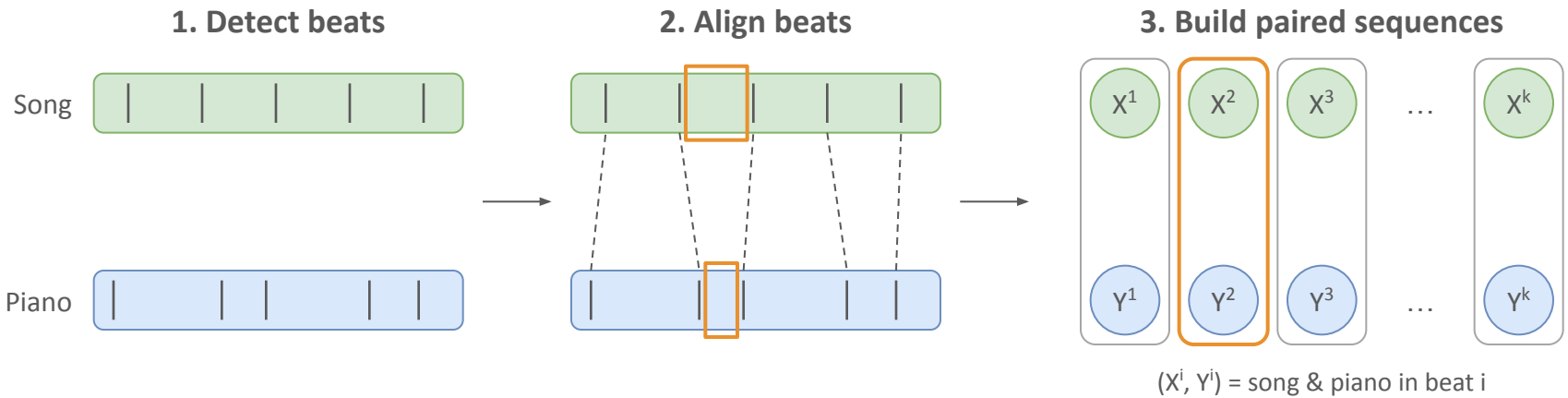
PiCoGen2

We aim to build an end-to-end piano cover generation system.

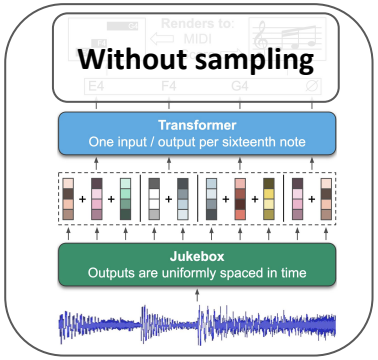


PiCoGen2 - data

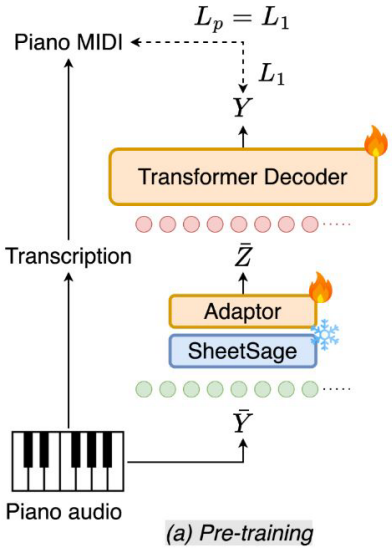
Training data is synchronized based on the musical **beat** (quantized time), rather than raw **time** values.



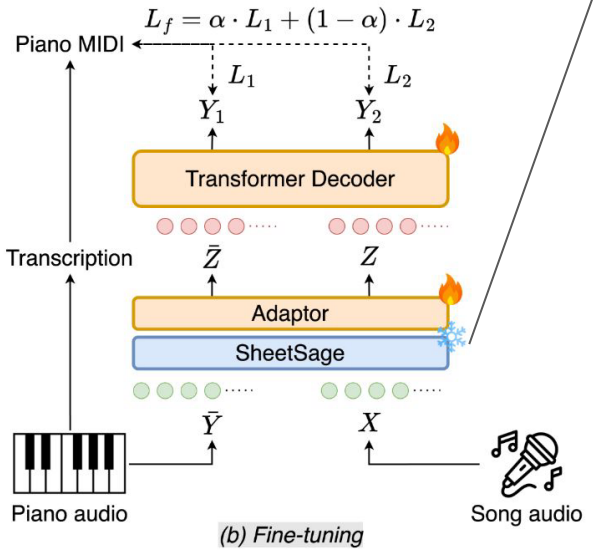
PiCoGen2 - training & inference



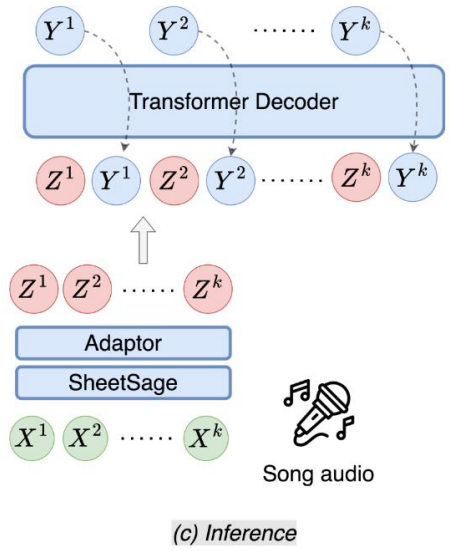
SheetSage



(a) Pre-training



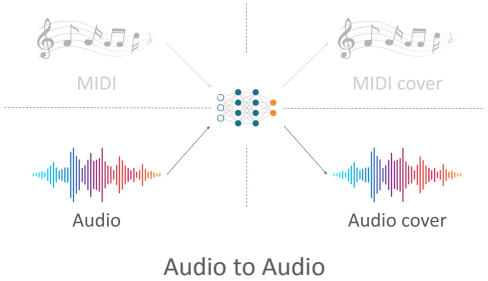
(b) Fine-tuning

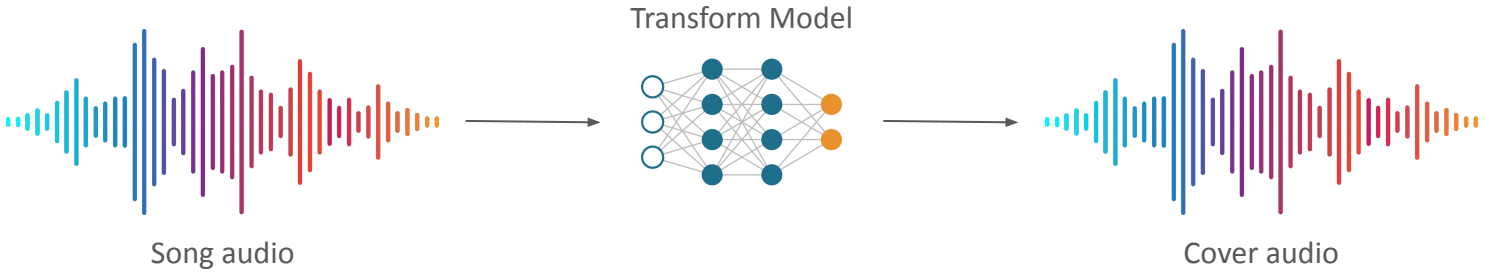


(c) Inference

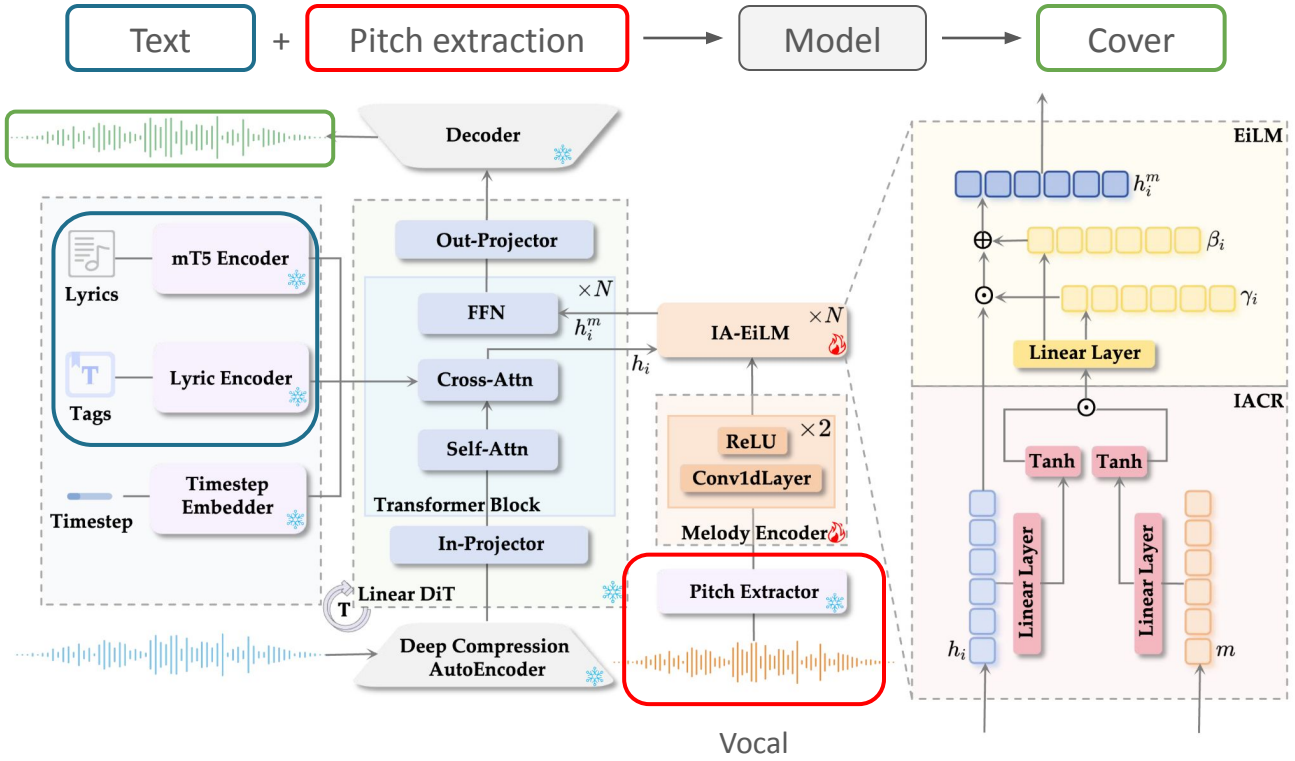
PiCoGen2 - demo

Task: audio-domain cover generation





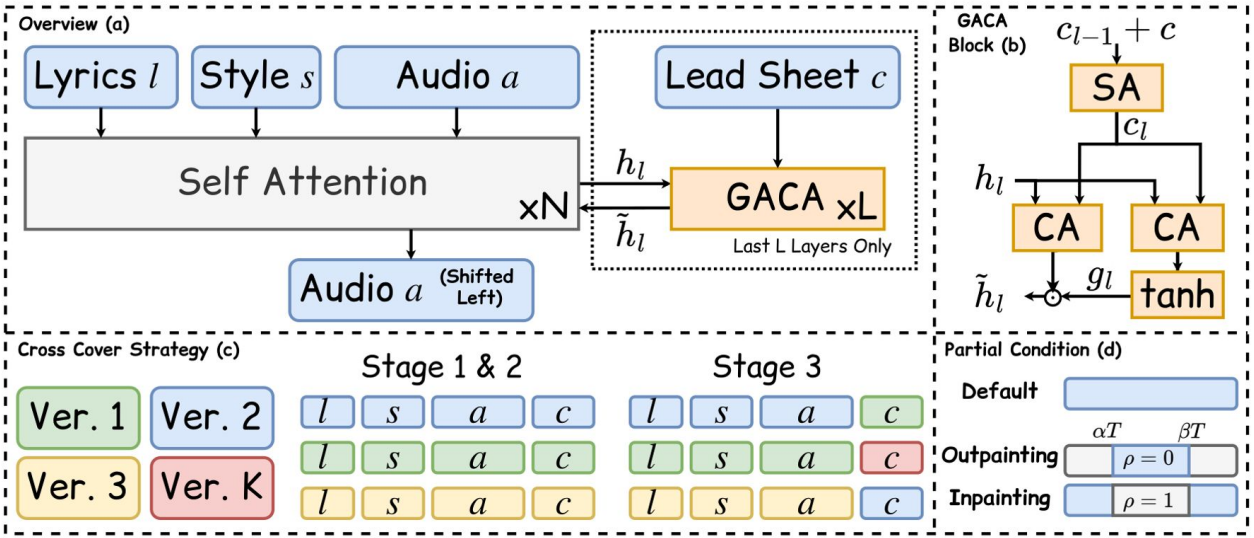
SongEcho



Li et al. “SongEcho: Towards Cover Song Generation via Instance-Adaptive Element-wise Linear Modulation” in Proc. of ICLR 2026

MuLaCover

Goal: flexible cover — free tempo, structure, and partial reference (no frame-level alignment)



Stage 1 · Short-clip warmup
30 s clips; learn lead sheet → audio

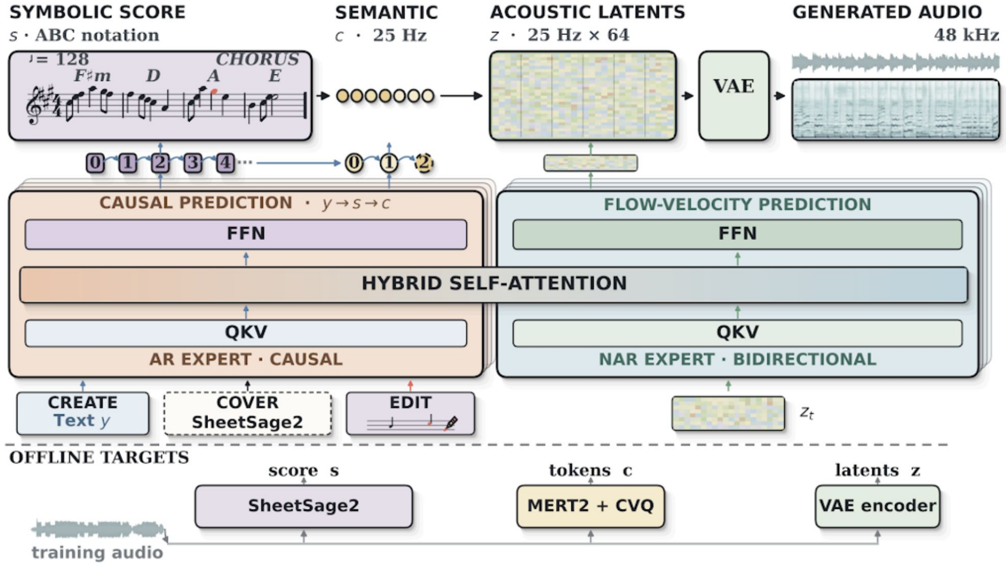
Stage 2 · Full-song training
Full songs; all inputs from one version

Stage 3 · Cross-cover finetuning
Lead sheet from another version (+ LoRA)

Yi et al. "MuLaCover: Flexible Cover Song Generation via Symbolic Tonal Control" in arXiv 2026

YuE2

Idea: compose in symbols, perform in audio. The score (melody + chords) is an explicit, editable plan.



Create
Text & lyrics ⟶ score ⟶ song

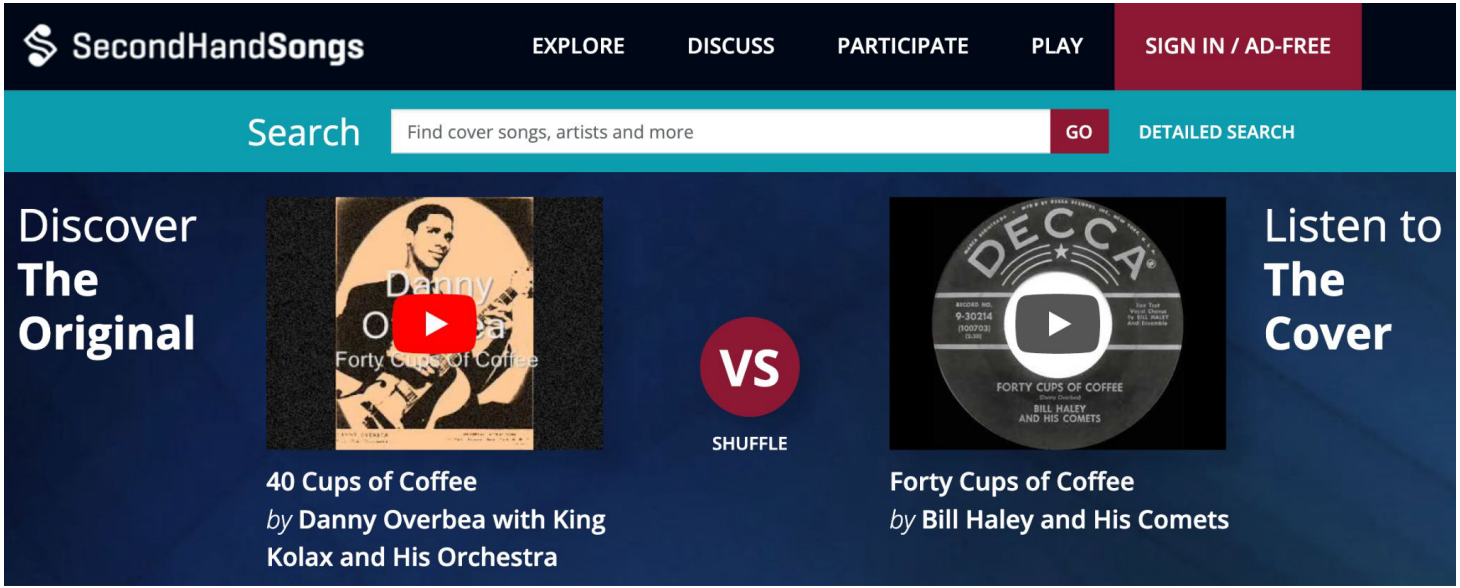
Cover (zero-shot)
Transcribe the source score ⟶ new style

Edit
Edit the score ⟶ re-render the song

Yuan et al. "YuE2: Unifying Symbolic and Audio Music Generation at Frontier Quality" in arXiv 2026

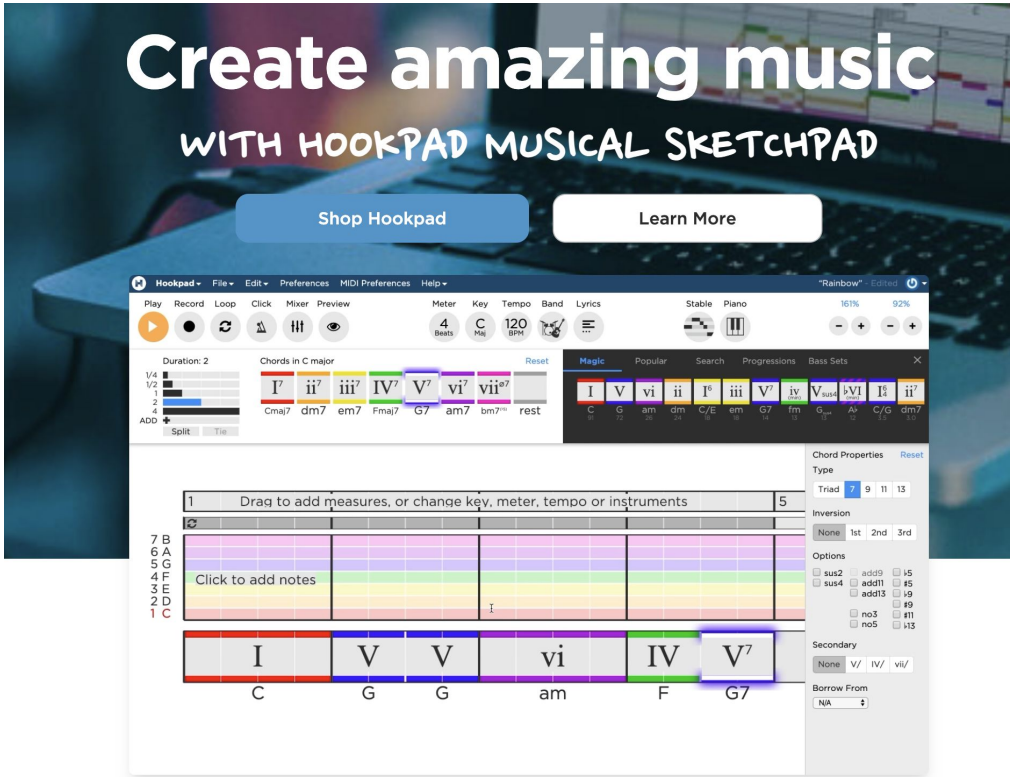
Resources

SecondHandSongs



https://secondhandsongs.com/

Hooktheory



https://www.hooktheory.com/

Datasets

Dataset	Type	Size	Content
POP909 (2020)	MIDI	909 songs	Piano arrangements + beat, chord, key
Live Orchestral Piano (2017)	MIDI pairs	223 pairs	Piano scores ↔ orchestrations
Pop2Piano (2023)	Audio pairs	4,989 pairs (307 h)	Pop songs ↔ synced piano covers
Discogs-VI (2024)	Metadata + audio links	1.9M versions, 348k cliques	Version cliques; 493k on YouTube
DiVers (2026)	Metadata + audio links	1.1M+ versions	In-the-wild versions (live, amateur)

If you are interested in doing anything related to cover song generation, feel free to contact me

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